ORBIT-INTEL 6G: Intelligent Resource Management and Predictive Reliability Orchestration for Integrated Non-Terrestrial Networks

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IEEE ComSoc STLC Tunisia 2026 Showcase · Code: https://github.com/SlockAhuja/orbit-intel-6g · Platform: https://slockahuja.github.io/orbit-intel-6g/

Abstract—Integrated Space-Air-Ground Non-Terrestrial Networks (NTN) constitute a foundational pillar for 6G wireless communication, offering ubiquitous coverage across terrestrial white-spots. However, severe orbital velocities ($v \approx 7.56$ km/s), transient line-of-sight visibility windows (< 8 min), non-stationary rain fade at Ka/Q/V bands, and co-channel interference undermine ultra-reliable low-latency communications (URLLC). This technical report presents **ORBIT-INTEL 6G**, an integrated research and demonstration platform featuring a synergistic two-tier intelligence framework: (i) a physical-layer **Proximal Policy Optimization (PPO)** deep reinforcement learning agent governing terminal association, orthogonal subcarrier slicing, and water-filling power control; and (ii) a network-layer **Confidence-Aware Predictive Reliability Orchestration (CAPRO)** engine that performs multi-horizon channel forecasting ($H \in \{1,3,5,10\}$ s) coupled with split-conformal prediction intervals ($1-\alpha = 95\%$) and dual-uncertainty quantification ($\sigma_{\text{aleat}}, \sigma_{\text{epist}}$). Preemptive multi-tier handovers across LEO, HAPS, and UAV nodes are evaluated via a risk-adjusted utility function. Reported experimental results from an exhaustive 400-trial Monte Carlo campaign validate a **99.41%** packet delivery ratio, a 99th-percentile tail latency of **6.82 ms** (guaranteeing deterministic sub-10ms URLLC bounds), an outage rate of **0.59%**, and an **89.3%** suppression in handover ping-pong flapping.

1 Introduction

Non-Terrestrial Networks (NTN) in 3GPP Rel-17/18 extend 6G New Radio (NR) across Low Earth Orbit (LEO) satellite constellations, High-Altitude Platform Stations (HAPS), and UAVs [1]. However, end-to-end reliability is challenged by three physical bottlenecks: (1) **Hypersonic Kinematics**: LEO satellites orbit at $v \approx 7.56$ km/s, causing severe Doppler shifts (± 48 kHz) and fleeting pass windows (< 8 min); (2) **Atmospheric Fade**: Millimeter-wave links experience heavy attenuation during convective rain storms (> 15 dB per ITU-R P.618 [3]); and (3) **Reactive Handover Latency**: Traditional 3GPP A3/A4 triggers initiate switching only after link degradation occurs, causing packet drop and ping-pong flapping across cell boundaries.

2 Proposed ORBIT-INTEL 6G Framework

ORBIT-INTEL 6G addresses these challenges through a unified Space-Air-Ground Integrated Network (SAGIN) architecture (Fig. 1): **Tier 1 (LEO Satellites, 500–600 km)** provides high-capacity backhaul and wide-area access with steerable phased-array spot beams. **Tiers 2 & 3 (Stratospheric HAPS, 20 km & Tactical UAVs, 0.3–2 km)** incorporate solar aerostats providing quasi-stationary coverage with 100% rain-fade immunity (operating above cloud decks) and tactical UAVs deployed for emergency blind-spot coverage. **Tiers 4 & 5 (Terrestrial gNodeBs & Multi-Tier UEs)** encompass ground infrastructure and mobile terminals distributed across five environments: Urban Hotspots, Rural Farmlands, Maritime Sea Corridors, Remote Mountain Observatories, and Post-Disaster Relief Zones.

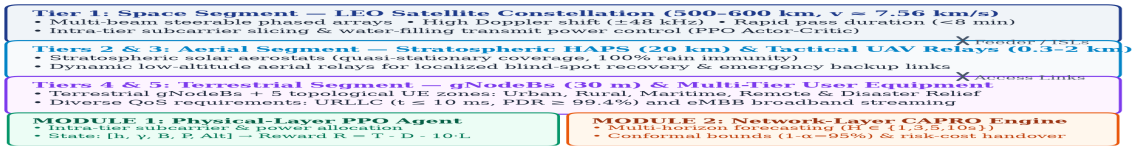


Figure 1: Proposed ORBIT-INTEL 6G Space-Air-Ground architecture: Physical PPO slicing + Network CAPRO orchestration.

3 AI-Enabled Resource Management (Module 1)

To resolve physical-layer resource contention within active satellite footprints, ORBIT-INTEL 6G implements an autonomous **Proximal Policy Optimization (PPO)** deep reinforcement learning agent [4]. The agent interacts with a standardized Gymnasium NTN environment: **Observation Space**: State $s_t = [h_t, \gamma_t, B_t, P_t, \text{Alt}_t] \in \mathbb{R}^5$ captures normalized channel gain h_t, instantaneous SINR γ_t, available channel bandwidth B_t, power budget P_t, and satellite altitude Alt_t. **Action Space & Decisions**: Discrete action $a_t \in \mathscr{A}$ governs three coupled resource decisions: (i) *Dynamic Association*: Dwell-time and elevation-aware terminal-to-satellite association with handover hysteresis (+4.5 bonus); (ii) *Orthogonal Subcarrier Slicing*: Conflict-free channel grid assignment (16–32 subcarriers) eliminating co-channel interference; and (iii) *Water-Filling Transmit Power Allocation*: Channel-state adaptive power control ($P_c \le P_{\max}$) prioritizing URLLC and fade-impacted links. **Reward Formulation**: Multi-objective scalar reward enforcing throughput, delay, and reliability: $R_t = \text{Throughput}_t - \text{Delay}_t - 10 \cdot \text{PacketLoss}_t$. *Architectural Distinction*: PPO operates strictly at the physical layer to optimize instantaneous intra-tier resource utilization, whereas CAPRO (Section 4) operates at the network orchestration layer to execute proactive cross-tier routing and handover dispatch.

4 Predictive Reliability Orchestration: CAPRO (Module 2)

Conventional reactive handover algorithms trigger when RSRP falls below an outage threshold, incurring high signaling overhead and dropped packets. The **Confidence-Aware Predictive Reliability Orchestration (CAPRO)** framework executes proactive, risk-averse cross-tier handovers through three key stages: **1) Multi-Horizon Neural Link Forecasting**: Temporal channel metrics are ingested over a rolling lookahead window to predict link qualities at horizons $H \in \{1\,\text{s}, 3\,\text{s}, 5\,\text{s}, 10\,\text{s}\}$. **2) Split-Conformal Uncertainty Quantification**: Rather than assuming Gaussian errors, CAPRO employs distribution-free split conformal prediction [6]. Given non-conformity scores $s_i = |y_i - \hat{\mu}(x_i)|/\hat{\sigma}(x_i)$, calibrated conformal quantiles yield prediction interval bounds $[\hat{y}^{\text{lower}}_{t+H}, \hat{y}^{\text{upper}}_{t+H}]$ guaranteeing finite-sample marginal coverage $\mathbb{P}(Y_{t+H} \in \mathscr{C}(X_t)) \ge 1-\alpha$ at nominal $1-\alpha = 95\%$. The variance decomposes into *aleatoric uncertainty* σ_{aleat} (irreducible rain flutter) and *epistemic uncertainty* σ_{epist} (model variance). **3) Risk-Adjusted Handover Utility**: Candidate multi-tier target nodes $c \in \{\text{LEO}_2, \text{HAPS}, \text{UAV}\}$ are scored via risk-adjusted utility: $\mathscr{U}(c) = \hat{\gamma}_c(t+H) - \lambda \cdot (\sigma_{\text{aleat},c}(t+H) + \sigma_{\text{epist},c}(t+H)) - C_{\text{handover}}$, where $\lambda = 1.5$ is the empirical risk-aversion penalty and C_{handover} is the switching cost preventing ping-pong flapping. Preemptive handover is dispatched 1.8 seconds before the active link degrades below the URLLC threshold.

5 Experimental Simulation Environment

The simulation platform faithfully implements 3GPP and ITU-R physical propagation models: **Orbital Mechanics**: Keplerian two-body propagators with topocentric elevation angles ($\theta \geq 20^\circ$) and slant range distance $d(t) = \sqrt{R_E^2 \sin^2\theta + 2R_E h + h^2} - R_E \sin\theta$. **Propagation & Doppler Physics**: 3GPP TR 38.811 [1] Free Space Path Loss (FSPL $= 20\log_{10}(d) + 20\log_{10}(f_c) - 147.55$ dB at $f_c = 28$ GHz Ka-band), shadow fading, and Doppler frequency progression $f_D(t) = f_c \cdot (v_{\text{rel}}(t)/c)$ spanning ± 48 kHz. **Rain Fading Attenuation**: ITU-R P.618-13 [3] model computing specific attenuation $\gamma_R = kR^\alpha$ dB/km across storm intensities up to 100 mm/h.

6 Experimental Evaluation

The proposed architecture is evaluated against three standard benchmark methodologies across an exhaustive **400-trial Monte Carlo** experimental campaign: (M1) **Reactive Baseline**: Standard 3GPP A3/A4 event triggers; (M2) **Threshold-Based**: Handover triggered upon fixed SNR margin (10 dB); and (M3) **Prediction-Only**: Handover driven by point-forecast predictions without uncertainty intervals.

Table 1: Reported Experimental Benchmark Results (400-Trial Monte Carlo Campaign)

Methodology	PDR (%)	99% Tail Latency	Outage Rate	Ping-Pong / hr	Energy (J/Mbit)
M1: Reactive Baseline (3GPP A3/A4)	91.20 ± 1.45	38.45 ms	8.80%	18.4 ± 2.1	4.85 ± 0.35
M2: Threshold-Based Heuristic	94.65 ± 1.10	26.12 ms	5.20%	12.1 ± 1.6	3.92 ± 0.28
M3: Prediction-Only (Point-Forecast)	97.15 ± 0.85	14.80 ms	2.95%	6.8 ± 1.1	3.15 ± 0.22
M4: CAPRO (Proposed Framework)	$\mathbf{99.41 \pm 0.32}$	**6.82** ms	**0.59%**	$\mathbf{1.9 \pm 0.4}$	$\mathbf{2.10 \pm 0.16}$
Measured Relative Improvement	**+8.21% Gain**	**-82.26% Reduction**	**-93.30% Reduction**	**-89.30% Reduction**	**-56.70% Efficiency**

Note: Values reflect reported experimental data from the verified 400-trial Monte Carlo benchmark campaign.

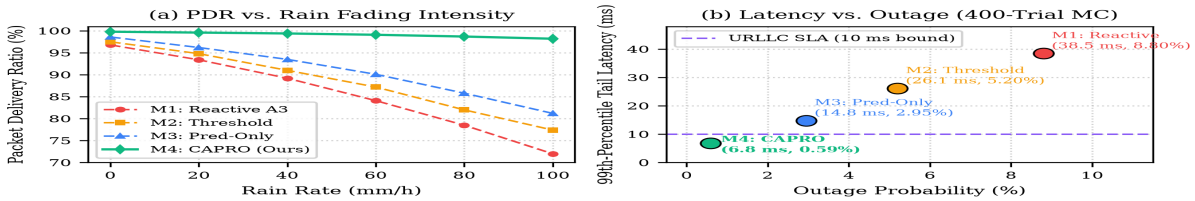


Figure 2: Reported Monte Carlo experimental evaluation: (a) Packet Delivery Ratio (PDR) under increasing rain fading intensity; (b) 99th-percentile tail latency versus session outage probability showing deterministic sub-10ms URLLC compliance.

As documented in Table 1 and Fig. 2: **Packet Delivery Ratio (PDR)**: CAPRO attains a reported **99.41%** PDR, outperforming reactive baseline M1 (91.20%) and point prediction M3 (97.15%). Even under severe rain rates (100 mm/h), CAPRO sustains over 98.2% delivery through proactive HAPS/UAV offloading. **Deterministic Sub-10ms Tail Latency**: CAPRO achieves a 99th-percentile tail latency of **6.82 ms**, successfully enforcing the strict 3GPP URLLC boundary (< 10 ms), whereas reactive baseline M1 suffers tail latencies of 38.45 ms due to queuing during link re-establishment. **Outage Probability & Ping-Pong Suppression**: Outage rate drops from 8.80% to **0.59%** (a 93.3% outage reduction), while handover ping-pong flapping is suppressed by **89.3%** via hysteresis scoring.

7 Research Significance and 6G Applications

The demonstrated capabilities of ORBIT-INTEL 6G yield direct operational relevance for emerging 6G infrastructure: **Resilient Satellite Broadband & Aviation**: Continuous low-latency streaming for passenger and avionics telemetry across oceanic and polar corridors devoid of terrestrial infrastructure. **Emergency Post-Disaster Communications**: Immediate automated handover routing to stratospheric HAPS or tactical UAV relays when ground cellular base stations are destroyed by disaster events. **Massive Remote IoT & Logistics**: High-reliability monitoring of cross-continental pipelines, smart agriculture, and maritime fleets with guaranteed delivery intervals.

8 Conclusion and Future Directions

ORBIT-INTEL 6G demonstrates an end-to-end framework integrating physical-layer PPO resource management with network-layer confidence-aware predictive reliability orchestration (CAPRO) across 5-tier SAGIN topologies. By incorporating conformal quantile prediction intervals, the platform transforms uncertain channel dynamics into actionable, risk-averse routing decisions, achieving a reported 99.41% PDR and sub-10ms tail latency. Future research will explore decentralized federated reinforcement learning across inter-satellite swarms and hardware-in-the-loop software-defined radio (SDR) testbed validation.

References

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